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1. Comparison: SHA-1 vs. MD5

Both are cryptographic hash functions used to verify data integrity, but they differ significantly in security and structure.

No.	Feature	MD5 (Message Digest 5)	SHA-1 (Secure Hash Algorithm 1)
1	**Full Form**	Message Digest 5.	Secure Hash Algorithm 1.
2	**Digest Length**	Produces a **128-bit** (16 bytes) hash value.	Produces a **160-bit** (20 bytes) hash value.
3	**Speed**	Faster and simpler to compute.	Slower than MD5 due to increased complexity.
4	**Security**	Highly vulnerable to collision attacks; considered cryptographically broken.	More secure than MD5, but now also considered vulnerable to sophisticated attacks.
5	**Complexity**	Simple design with 4 rounds of 16 steps each.	More complex design with 4 rounds of 20 steps each (80 rounds total).
6	**Brute Force**	Requires 2^{128} operations to find the original message from a digest.	Requires 2^{160} operations, making it harder to crack via brute force.
7	**Collision**	Requires 2^{64} operations to find two different messages	Requires 2^{80} operations to find two messages with the

	Resistance	with the same hash.	same hash.
8	**Release Year**	Developed by Ronald Rivest in **1991**.	Developed by the NSA and published in **1995**.

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2. Comparison: AES vs. DES

These are symmetric-key block ciphers used for data encryption.

No.	**Feature**	**DES (Data Encryption Standard)**	**AES (Advanced Encryption Standard)**
1	**Block Size**	Fixed block size of **64 bits**.	Fixed block size of **128 bits**.
2	**Key Length**	Effective key length is **56 bits** (8 bits are parity).	Variable key lengths: **128, 192, or 256 bits**.
3	**Structure**	Based on the **Feistel Network** structure.	Based on **Substitution-Permutation Network**.
4	**Security**	Insecure for modern use; can be brute-forced in hours.	Highly secure; currently the global standard for data encryption.
5	**Speed**	Relatively slow (especially in software).	Much faster than DES in both hardware and software.
6	**Number of**	Always involves **16 rounds**.	Number of rounds depends on

	Rounds		key size: **10, 12, or 14**.
7	**Design Origin**	Developed by IBM in **1975**.	Developed by Vincent Rijmen and Joan Daemen in **2001**.
8	**Software/Hardware**	Better suited for hardware implementation.	Efficiently implemented on both hardware and software platforms.

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3. Difference: Monoalphabetic vs. Polyalphabetic Substitution

These are traditional symmetric cipher techniques where plaintext characters are replaced by ciphertext.

No.	**Feature**	**Monoalphabetic Substitution**	**Polyalphabetic Substitution**
1	**Mapping**	**One-to-one**: Each plaintext character is always replaced by the same ciphertext character.	**One-to-many**: A single plaintext character can map to different ciphertext characters.
2	**Number of Alphabets**	Uses a **single** substitution alphabet for the entire message.	Uses **multiple** substitution alphabets at different positions.
3	**Security Level**	Low; easily broken using frequency analysis.	High; resistant to simple frequency analysis.
4	**Complexity**	Simple to implement and understand.	Complex and harder to implement manually.

5	**Key Dependence**	The key is independent of the character's position in the text.	The substitution depends on both the character and its **position**.
6	**Examples**	Caesar Cipher, Atbash Cipher, Additive Cipher.	Vigenère Cipher, Hill Cipher, Playfair Cipher, Enigma.
7	**Frequency Analysis**	Frequency of letters in ciphertext reflects the frequency in plaintext.	Frequency of letters is "flattened," making patterns harder to find.
8	**Relationship**	Static relationship between Plaintext and Ciphertext.	Dynamic relationship based on a keyword or key-stream.